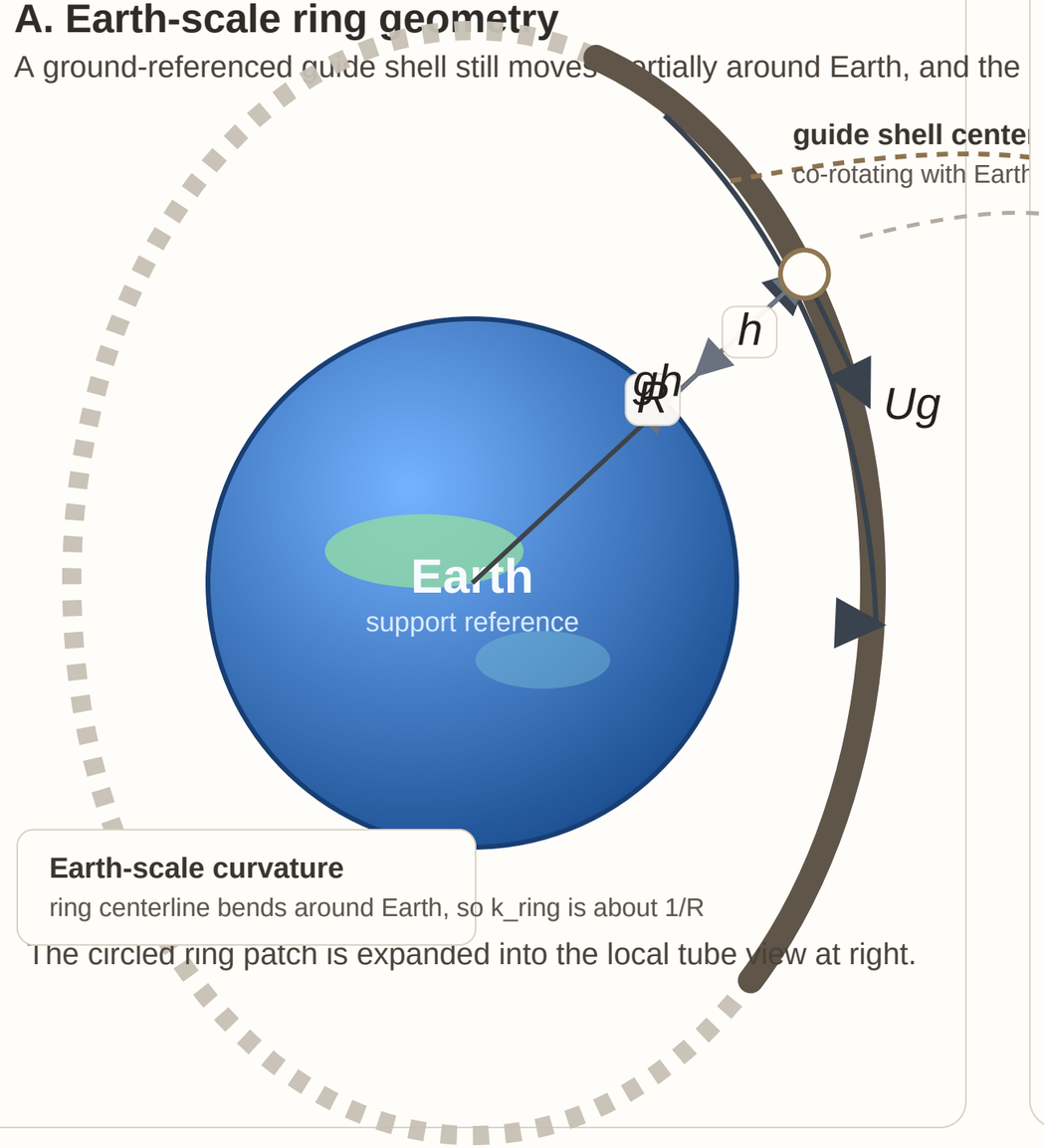
Global ring geometry and local coordinate convention

Figure 1. Earth scale at left, local torus patch at right (conceptual, not to scale)

A. Earth-scale ring geometry

A ground-referenced guide shell still moves partially around Earth, and the

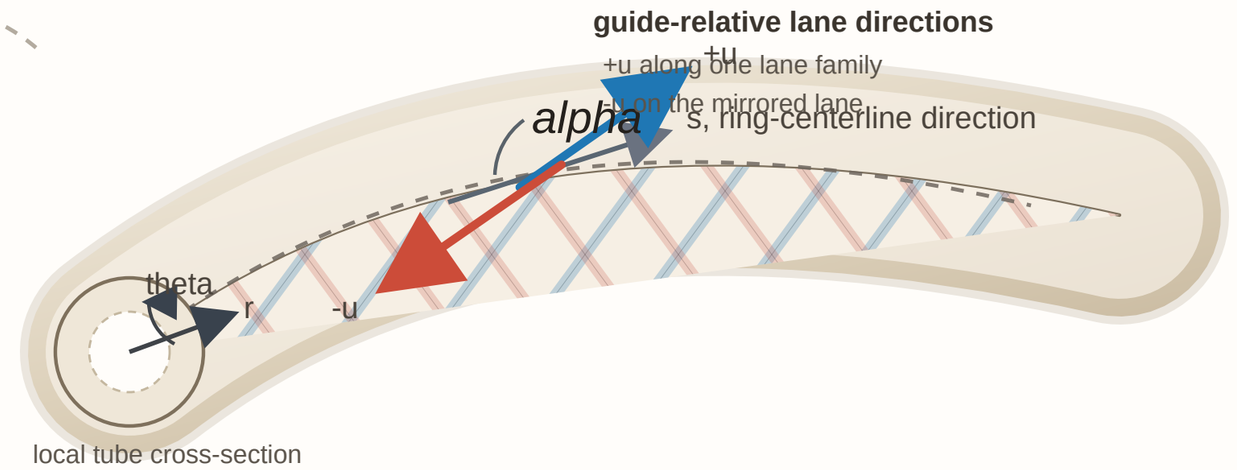


Earth-scale curvature
ring centerline bends around Earth, so k_ring is about 1/R

The circled ring patch is expanded into the local tube view at right.

B. Local torus-patch coordinates

The local bookkeeping uses ring-centerline direction s, azimuth theta around the tube, radial direction r,



Local coordinate convention
s is distance along the ring centerline around Earth.
theta is azimuth around the torus cross-section, and r is local radial direction normal to that cross-section.

helix angle
alpha is measured relative to the local s directio